HAO ZOU

(631) 809-8931 ◇ hz2999@columbia.edu ◇ Website ◇ Scholar

EDUCATION

Columbia University — New York, NY
M.S. in Computer Science, Machine Learning Track — *09/2024 – 05/2026*

University of Minnesota, Twin Cities — Minneapolis, MN
B.S. in Computer Science — *09/2019 – 05/2023*
Research Interests: Reinforcement Learning, LLM Agents, Agentic Search, Reliable Generation, LLM Reasoning, Diffusion Language Models

SELECTED PUBLICATIONS

Detect, Remask, Repair: Diffusion Editing for Faithful Summarization of Evolving Contexts
Hao Zou, Zachary Horvitz, Chandhru Karthick, Zhou Yu, Kathleen McKeown
EMNLP 2026 Main Conference, Oral Presentation. [paper] [arXiv]

OpenForgeRL: Train Harness-native Agents in Any Environment
Xiao Yu, Baolin Peng, Ruize Xu, **Hao Zou**, Qianhui Wu, Hao Cheng, Wenlin Yao, Nikhil Singh, Zhou Yu, Jianfeng Gao
ICLR 2027 submission. [paper]

OSWorld 2.0: Benchmarking Computer Use Agents on Long-Horizon Real-World Tasks
Mengqi Yuan, Zilong Zhou, Xinzhuang Xiong, . . . , Xiao Yu, **Hao Zou**, . . . , Tao Yu
ICLR 2027 submission. [project] [arXiv]

Rethinking Reasoning with MDLMs: Early Exits, Post-hoc Reasoning, and Beyond
Zachary Horvitz, Raghav Singhal, **Hao Zou**, Carles Domingo-Enrich, Zhou Yu, Rajesh Ranganath, Kathleen McKeown
COLM 2026. [arXiv]

You Make me Feel like a Natural Question: Training QA Systems on Transformed Trivia Questions
Tasnim Kabir, Yoo Yeon Sung, Saptarashmi Bandyopadhyay, **Hao Zou**, Abhranil Chandra, Jordan Boyd-Graber
EMNLP 2024. [paper]

A Survey of Diffusion Models in Natural Language Processing
Hao Zou, Zae Myung Kim, Dongyeop Kang
arXiv, 2023. [paper]

Debiasing Language Models for In-Context Learning Using a Causal Inference-Inspired Method
Hao Zou, Karin de Langis, Dongyeop Kang, Yohan Jo
Manuscript, 2023. [paper]

RESEARCH EXPERIENCE

NLP Lab / Department of Computer Science, Columbia University — 02/2025 – Present
Research Staff Associate II / Graduate Researcher — *Advisors: Kathleen McKeown & Zhou Yu*

- **Faithful summarization and IARPA BENGAL:** First-authored *Detect, Remask, Repair* (EMNLP 2026), a masked-diffusion framework that detects unsupported spans, selectively remasks them, and repairs only unreliable content. Introduced StreamSum for evolving-context summarization; one-step repair reduces localized revision latency to under 0.5s.
- Lead evaluation and reproducibility work for an IARPA-funded faithful-summarization effort, including metric pipelines, result analysis, containerized evaluation, and integration of repair, contamination detection, and QA components for an interactive demo; extending the pipeline to multilingual Arabic-dialect summarization.
- **Masked-diffusion reasoning:** Co-developed reasoning-as-infilling, answer-conditioned posterior reasoning, uncertainty-aware early exit, and intermediate reasoning validation for MDLMs (COLM 2026). Posterior-trace fine-tuning improves LLaDA-8B-Base GSM8K accuracy by 14.9 points.

- **LLM agents:** Contributed long-horizon computer-use workflows to OSWorld 2.0, a 108-task benchmark with 31 self-hosted websites. Co-authored OpenForgeRL, a harness-native framework connecting real agent harnesses and isolated rollout environments to scalable SFT/RL training.
- Investigating RL for long-horizon GUI and source-grounded web-search agents, including asynchronous rollouts, tool-use credit assignment, and fresh-news multi-hop retrieval.

Duke University 05/2024 – 01/2025
Graduate Research Assistant *Advisor: Enmao Diao*

- Built benchmarking pipelines comparing diffusion and flow-matching training paradigms, prediction objectives, and architectures; explored a triangle-distribution-based objective for improved training stability and generation quality.

NLP Lab, University of Minnesota – Twin Cities 08/2021 – 05/2023
Undergraduate Research Assistant *Advisor: Dongyeop Kang*

- Developed a causal-inference-inspired method for estimating the effect of input text on candidate labels, improving in-context learning accuracy by up to 22 percentage points across evaluated tasks.
- First-authored a survey of diffusion models for NLP covering diffusion formulations, denoising objectives, non-autoregressive generation, token-level control, robustness, and scaling directions for large diffusion language models.

CLIP Lab, University of Maryland 06/2021 – 06/2022
Undergraduate Research Assistant *Advisor: Jordan Boyd-Graber*

- Developed question-rewriting and DPR retrieval pipelines for open-domain QA, improving cross-domain accuracy by 18% in the evaluated setting; analyzed 2K+ generated questions across domain and syntactic shifts.

WORK EXPERIENCE

IBM 10/2024 – 01/2025
AI Algorithms Intern – LLM Efficiency / Quantization *Mentor: Naigang Wang*

- Proposed a Johnson–Lindenstrauss-based KV-cache quantization approach integrated with sparsity pruning, reducing memory usage by 50% and improving inference speed by 2× in the evaluated setup; benchmarked quantized attention across multiple LLM families.

Sony Research 01/2022 – 05/2023
Research Intern – Reinforcement Learning for Language Generation

- Developed a multi-style controllable text-generation pipeline using PPO and representation-fusion methods, improving human-judged style alignment by 31.2%.

SKILLS & SERVICE

Research: Masked diffusion LMs, faithful summarization, LLM reasoning, LLM agents, agentic search, evaluation/verifiers
RL / ML Systems: PPO, GRPO, SFT/RL, PyTorch, Hugging Face, vLLM, veRL, Ray, FSDP, LoRA, CUDA
Engineering: Python, C/C++, Linux, Git, Docker, Kubernetes; long-horizon rollouts, GUI/tool-use agents
Academic Service: Reviewer for ACL, EMNLP, EACL